

Explain and evaluate multi-step solutions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid? Copy or complete the first step.

2. A solution to $3,840 / 24 \times 15$ gives 10. Find the error and correct it.

3. Solve $96 \times 45 + 8,640 / 32$.

4. Miro says: Miro assumes every number in a word problem must be used. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid? Copy or complete the first step.

Answer Multiplication must represent the total books first, so multiply first. The answer is 2,240.

2. A solution to $3,840 / 24 \times 15$ gives 10. Find the error and correct it.

Answer $3,840 / 24 = 160$, then $160 \times 15 = 2,400$.

3. Solve $96 \times 45 + 8,640 / 32$.

Answer 4,590.

4. Miro says: Miro assumes every number in a word problem must be used. Is this correct? Explain.

Answer Use only information that changes the quantity asked for.

Explain and evaluate multi-step solutions

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?

6. Check this result using an inverse, estimate or known fact: Solve $96 \times 45 + 8,640 \div 32$.

2. A solution to $3,840 \div 24 \times 15$ gives 10. Find the error and correct it.

7. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.

3. Solve $96 \times 45 + 8,640 \div 32$.

8. Identify and correct this misconception: Miro assumes every number in a word problem must be used.

4. Solve this independently and show every stage: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A solution to $3,840 \div 24 \times 15$ gives 10. Find the error and correct it.

10. Write and solve a similar question to: Solve $96 \times 45 + 8,640 \div 32$.

Teacher answers and checking prompts

1. Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?
Answer Multiplication must represent the total books first, so multiply first. The answer is 2,240.
2. A solution to $3,840 / 24 \times 15$ gives 10. Find the error and correct it.
Answer $3,840 / 24 = 160$, then $160 \times 15 = 2,400$.
3. Solve $96 \times 45 + 8,640 / 32$.
Answer 4,590.
4. Solve this independently and show every stage: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?
Answer Multiplication must represent the total books first, so multiply first. The answer is 2,240.
5. Use a second method or representation for: A solution to $3,840 / 24 \times 15$ gives 10. Find the error and correct it.
Answer Expected result: $3,840 / 24 = 160$, then $160 \times 15 = 2,400$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Solve $96 \times 45 + 8,640 / 32$.
Answer Expected result: 4,590. A valid check must be shown.
7. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.
Answer Answers vary. The unused information must not affect the required calculation.
8. Identify and correct this misconception: Miro assumes every number in a word problem must be used.
Answer Use only information that changes the quantity asked for.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Solve $96 \times 45 + 8,640 / 32$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Explain and evaluate multi-step solutions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.

6. Create a new example that tests the same idea as: Solve $96 \times 45 + 8,640 \div 32$.

2. Prove the answer to this question, rather than only calculating it: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A solution to $3,840 \div 24 \times 15$ gives 10. Find the error and correct it.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 4,590.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro assumes every number in a word problem must be used.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.
Answer Answers vary. The unused information must not affect the required calculation.
2. Prove the answer to this question, rather than only calculating it: Two methods solve $48 \times 125 - 3,760$. One multiplies first; one subtracts first. Which is valid?
Answer Expected result: Multiplication must represent the total books first, so multiply first. The answer is 2,240. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A solution to $3,840 / 24 \times 15$ gives 10. Find the error and correct it.
Answer Expected result: $3,840 / 24 = 160$, then $160 \times 15 = 2,400$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 4,590.
Answer One valid reconstruction is: Solve $96 \times 45 + 8,640 / 32$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro assumes every number in a word problem must be used.
Answer Use only information that changes the quantity asked for.
6. Create a new example that tests the same idea as: Solve $96 \times 45 + 8,640 / 32$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.